

Programming for Business

**LO1: Introduction to Computers &
Programming
FAST–NUCES**

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Session Roadmap (LO1)

- Computers basics
- Hardware & software
- Data storage
- Programs & logic
- Excel & Python examples

Question

Which business CANNOT run without computers today?

- Bank
- Retail Store
- University
- All of them?

What is a Computer?

- Electronic machine
- Takes input
- Processes data
- Gives output

IPO Model (Very Important)

- Input → Process → Output
- Business Example:
- Sales Data → Calculation → Profit Report

Hardware – Overview

- Physical components
- You can touch them
- Needed to run software

Hardware Categories

- Input devices
- Processing (CPU)
- Memory (RAM)
- Storage (SSD/HDD)

Fun Activity – Hardware

Your laptop suddenly stops.

Is it:

- Hardware issue?
- Software issue?

Software – Overview

- Programs
- Instructions
- Tells hardware what to do

Types of Software

- System Software (Windows)
- Application Software (Excel, Browser)

Business Software Examples

- Excel – Reports
- ERP – Inventory
- CRM – Customers

How Computer Stores Data

- Data stored as 0s & 1s
- Binary system
- Stored in memory & disk

Data Size Concept

- Bit → Byte → KB → MB → GB
- Example: Excel file size

What is Programming?

- Giving instructions
- Step-by-step logic
- Automates business tasks

Program vs Human Work

Human:

- Slow
- Error-prone

Program:

- Fast
- Accurate

How a Program Works

- Input → Logic → Output
- Sales – Cost = Profit

Variables (Simple Idea)

- Store values
- Value can change

Sales = 50000

Excel as Programming Tool

- Cell = Variable
- Formula = Logic
- Sheet = Program

Excel Example

- Sales: 50,000
- Cost: 30,000
- Profit = Sales – Cost

Python Example (Conceptual)

- `sales = 50000`
- `cost = 30000`
- `profit = sales - cost`

Excel vs Python (Business)

Excel:

- Small data
- Reports

Python:

- Automation
- Large data

In-Class Exercise 1

- A shop sells products.
- What data is required to calculate profit?

Solution – Exercise 1

- • Sales
- • Cost
- • Profit

In-Class Exercise 2

- If sales > 100,000
- Give 10% discount
- Else no discount

Solution – Exercise 2

- IF sales > 100000
- Apply discount
- ELSE
- No discount

Real Business Use Cases

- Retail
- Banking
- HR
- E-commerce

Quick Recap (LO1)

- ✓ Computers
- ✓ Hardware & Software
- ✓ Programs
- ✓ Excel & Python